

# Timothy Do's Official CV

An A.I. Machine Vision Enthusiast & DSP Techie to Make the World A Better Place



## EDUCATION

### University of California Irvine, Irvine, CA — B.S. Electrical Engineering

September 2019 - June 2023 (Expected)

- 4th Year Undergraduate, Accelerated Status
- Overall GPA: 3.90, Major GPA: 3.91
- **Specialization:** Digital Signal Processing
- **Research Interests:** Digital Signal Processing with Artificial Intelligence, Digital Image Processing, Machine Vision
- **Relevant Coursework:**
  - EECS 22: Advanced C Programming, learned about advanced data structures, program decomposition (Winter 2021)
  - EECS 22L: Introduction to Software Engineering, worked in a team of 4 (TTLZ22s) to develop a network chess application in C with a git repo with version control (Spring 2021)
  - EECS 50/150: Discrete-Time/Continuous Signals and Systems, learned about LTI Systems, Z-Transforms, the DTFT, FFT, Anti-Aliasing, Laplace Transform (Spring 2021/Winter 2022)
  - EECS 152A/B: Digital Signal Processing, learned about the Fourier Transform, Sampling, Filter Design (Fall 2021)
  - EECS 203A: Digital Image Processing, learned about Image transforms, 2D Fourier Transform, MATLAB processing. (Spring 2022)
  - EECS 170A/B/C: Electronics, learned about semiconductor properties and the topologies of diodes, BJT, and MOSFET transistors and their applications
- **Major Coursework** for Fall 2022: EECS 159A (Senior Design Project), EECS 160A/LA (Control Systems), ENGR 190W (Engineering Writing)
- **Clubs and Societies:** IEEE-HKN (Eta Kappa Nu), Institute of Printed Circuits, Irvine Computer Vision Laboratory, Tau Beta Pi
- **Awards:** Dean's Honor List (GPA 3.5+) for 9 Quarters (Fall 2019, Winter 2020, Spring 2020, Fall 2020, Winter 2021, Spring 2021, Fall 2021, Winter 2022, Spring 2022)

### Evergreen Valley High School, San Jose, CA — Diploma

August 2015 - May 2019

- GPA: Unweighted: 3.8, Weighted: 4.0
- **Relevant Coursework:** AP Physics C: Mechanics (4), AP Computer Science A (4), AP Statistics (4), AP Calculus AB (5), AP US History (4), Math 072 (Calc II) @ Evergreen Valley College
- **Awards:** EVHS Band Scholar, Magna Cum Laude, Golden State Diploma, Bilingual Certification in Vietnamese, AP Scholar with Distinction

## CONTACT

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🔗 [Linkedin](https://www.linkedin.com/in/do-timothy/) (/in/do-timothy)

📁 [Github](https://github.com/dotimothy) (dotimothy)

<https://timothydo.me>

## SKILLS

### Computers:

Assembled/Debugged Desktop Hardware and OS installations.

### Electronics:

Soldered Circuits and Designed Components on Oscilloscopes, Power Supplies, Function Generator and Multimeter.

### Embedded System

#### Programming:

Programmed with C, Java, and Python libraries by creating IoT projects on Arduino/Raspberry Pis

### Image Processing & Machine Learning:

Programmed GPUs to do Image Processing in MATLAB & Tensorflow AI

### Web Socket Programming:

Created Websites with HTML, CSS, and Javascript in addition to TCP socket programming using Python.

## HACKATHONS

### Cisco Webex Hackathon 2021

[Gratition](#): An Easing Mental Health Web App (with Google Voice Synthesis)

**Languages:** HTML/CSS/JS

### ZotHacks 2021

[Anteatings](#): A Fun JS-Based Antsmashing Game on the Web

**Languages:** HTML/CSS/JS

# EXPERIENCE

## Western Digital, San Jose, CA - Industrial IOT Intern

June 2022 - September 2022

- **Skills:** MySQL, Python, Raspberry Pi, Soldering, Linux, Lean Six Sigma
- Certified as a Six Sigma Yellow Belt for learning Six Sigma foundations.
- Developed Presentation Skills in a Corporate Environment.
- Learned Industrial Tool Sets for Circuit Design and Embedded System Programming.

## UC Irvine, Irvine, CA - Student Researcher

January 2022 - Present

- Assisted in research under the direction of Professor Glenn Healey on the topic of Spectral Image Filtering for Emissions on Wildfires
- Digitized Imaging Filters into CSVs and compared them in MATLAB against certain chemical emissions.
- Generated altitude maps using QGIS & SRTM 1 Arc-Second Global

## IPC Club @ UCI, Irvine, CA - Vice President

November 2021 - Present

- Reserved club meeting rooms and sent weekly emails to club members regarding club events
- Programmed the club website and managed social media (i.e. moderator of club discord server)
- Led a Two-Part AM Radio Workshop where I had to design the circuit for the best amplifier gain in addition to creating easy-to-read slides & ordering parts from Amazon/Dollar Tree.

## Boy Scouts of America, San Jose, CA - Merit Badge Counselor

July 2021 - Present

- Offered to teach five Merit Badge Courses: Programming, Music, and Bugling, Electronics, and Digital Technology.

## UC Irvine D.C.E., Irvine, CA - I.T. General Assistant

July 2021 - June 2022

- Helped Set-Up Desktops, Docks, Webcams, and Computers for over 300+ Employees in UCI's Division of Continuing Education

## Environmental Injustice, San Jose, CA - Collaborator

June 2020 - September 2020

- Developed techniques such as stakeholder analysis, using Civic Data Tools such as CalEnviroScreen, and citing in Chicago format.

## Evergreen Music Mentors, San Jose, CA - Founding President

September 2018 - September 2021

- Lead music tutoring sessions for younger middle school students
- Integrated software (Cybernotes) as a video calling platform

## Most Holy Trinity Parish, San Jose, CA - Teacher's Assistant

October 2017 - May 2019

- Helped young students in their religious studies in Catholicism and their skills in Vietnamese.

## Boy Scouts of America Troop 610, San Jose, CA - Eagle Scout

September 2013 - September 2019

- Awarded the Bronze Palm: Earning 5+ additional Merit Badges over the 21 required for Eagle Scout.
- **Eagle Scout Project:** Constructing 2 Food Cabinets for the Milpitas Food Bank

# PUBLICATIONS

**Timothy Do;** Matthew Prata;

Jorge Radge; Alex Wang. 2022.

[Colorization Utilizing](#)

[Unsupervised Methods](#). UCI EECS

195, Network Science. UCI Henry

Samueli School of Engineering.

University of California, Irvine.

Tahis Alcantar; **Timothy Do;** April

Godinez; Daniel Jilani; Vicent

Marin; Wonhee Lee; Haoyang Liu;

Huiqi Mai; Andrew Ramirez; Jiaqi

Wu; Depei Xu. 2020. [Fast Disaster](#)

[Case Study: Santa Fe Springs](#). UCI

Anthropology 25A, Environmental

Injustice. Disaster STS Network.

University of California, Irvine.

Tahis Alcantar; **Timothy Do;** April

Godinez; Daniel Jilani; Vicent

Marin; Wonhee Lee; Haoyang Liu;

Huiqi Mai; Andrew Ramirez; Jiaqi

Wu; Depei Xu. 2020. [Slow](#)

[Disaster Case Study: Wilmington](#).

UCI Anthropology 25A,

Environmental Injustice. Disaster

STS Network. University of

California, Irvine.

Tahis Alcantar; **Timothy Do;** April

Godinez; Daniel Jilani; Vicent

Marin; Wonhee Lee; Haoyang Liu;

Huiqi Mai; Andrew Ramirez; Jiaqi

Wu; Depei Xu. 2020. [Combo](#)

[Disaster Case Study: San](#)

[Bernardino County](#). UCI

Anthropology 25A, Environmental

Injustice. Disaster STS Network.

University of California Irvine

# LANGUAGES

**English** (Fluent), **Vietnamese**

(Fluent), & **Spanish** (Proficient)

# PROJECTS

## [PhotoLabServer](#) - *A Web-Based Image Processing Server*

August 2022

- **Languages Used:** Python / MATLAB
- Implemented a Python Flask Web Server to process images
- Piped with HTTP Post Requests to send images, MATLAB to perform operation, then send images back with HTTP GET.
- Performs image operations like sharpen, smoothing, FFT.

## [TTLZNetChess](#) - *An Advanced Client-Based Chess Application*

March 2021 - June 2021

- Managed a team of four developers into develop a C-Based Chess Application for EECS22L (Intro to Software Engineering)
- Implemented Terminal Graphics (Colors) using Escape Codes
- Implemented AIs in Chess using Minimax Algorithm
- Networking Support using standard sockets and FCFS Protocol

## [RefreshVideos](#) — *A High Refresh Rate Video Player*

April 2020 - Present

- **Languages Used:** HTML / CSS / Javascript
- Used Canvas to play videos at high frame rates and apply effects
- Featuring filters like Edge Detection, Aging, etc.

## [pdDSP](#) — *Audio-Processing using Pure-Data*

March 2020

- **Languages Used:** Pure Data
- Used Pure Data to mix audio and designed audio filters (e.g. Low Pass Filter, Band Pass Filter)

## [PiFmMorse](#) — *A Morse Code FM Transmitter for the Raspberry Pi*

September 2020

- **Languages Used:** Python
- An Program to transmit Morse Code over FM radio via GPIO 4
- Learned to concatenate .wav files via wave library
- Learned to call linux commands using os library

## [RPiMorse](#) — *A Morse Code LED Circuit for the Raspberry Pi*

September 2020

- **Languages Used:** Python
- An Program to flash an LED in Morse Code given a circuit
- Learned GPIO Pins Wiring and its respective python library

## [SCU Complex Fires](#) — *A Website Providing Wildfire Resources*

August 2020 - September 2020

- **Languages Used:** HTML / CSS/ Javascript
- Provided Wildfire resources for research, shelter, and news
- Worked with a team of 4 and remotely collaborated through Github
- Features an Energy-Efficient Dark Mode